

The Sochocki Formula

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Abstract

The Sochocki formula (proven by Polish mathematician Julian Karol Sochocki) is stated and proven. This formula is also known as the “Dirac relation” in physics literature.

We want to derive the distribution identity

$$\frac{1}{x - i0^+} = \mathcal{P}\frac{1}{x} + i\pi\delta(x). \quad (0.1)$$

This should be understood in the sense of distributions, i.e. after integrating against a smooth test function $f(x)$. In particular we note that the principal value operator $\mathcal{P}$ is only meaningful when it acts on some integral.

We choose the minus sign in the expression $1/(x - i\eta)$ because it is equal to

$$\frac{x + i\eta}{x^2 + \eta^2}$$

where the numerator has a nice form.

1 Real part: principal value

Consider

$$\operatorname{Re} \frac{1}{x - i\eta} = \frac{x}{x^2 + \eta^2}, \quad \eta > 0. \quad (1.1)$$

We want to show

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} f(x) \frac{x}{x^2 + \eta^2} dx = \mathcal{P} \int_{-\infty}^{\infty} \frac{f(x)}{x} dx. \quad (1.2)$$

Write

$$f(x) = f(0) + xg(x), \quad g(x) = \frac{f(x) - f(0)}{x}, \quad (x \neq 0). \quad (1.3)$$

Then

$$\int f(x) \frac{x}{x^2 + \eta^2} dx = f(0) \int \frac{x}{x^2 + \eta^2} dx + \int g(x) \frac{x^2}{x^2 + \eta^2} dx, \quad (1.4)$$

$$= 0 + \int g(x) \frac{x^2}{x^2 + \eta^2} dx. \quad (1.5)$$

As $\eta \rightarrow 0$, $\frac{x^2}{x^2 + \eta^2} \rightarrow 1$, so by dominated convergence

$$\lim_{\eta \rightarrow 0^+} \int g(x) \frac{x^2}{x^2 + \eta^2} dx = \int g(x) dx. \quad (1.6)$$

Meanwhile,

$$\mathcal{P} \int \frac{f(x)}{x} dx = \int g(x) dx, \quad (1.7)$$

since the constant term vanishes in the principal value sense. Thus the real part gives the PV integral.

2 Imaginary part: delta function

Now consider

$$\operatorname{Im} \frac{1}{x - i\eta} = \frac{\eta}{x^2 + \eta^2}. \quad (2.1)$$

We compute

$$\int_{-\infty}^{\infty} f(x) \frac{\eta}{x^2 + \eta^2} dx. \quad (2.2)$$

Change variables $x = \eta t$:

$$\int_{-\infty}^{\infty} \frac{f(\eta t)}{1 + t^2} dt. \quad (2.3)$$

As $\eta \rightarrow 0$, $f(\eta t) \rightarrow f(0)$, so

$$\lim_{\eta \rightarrow 0^+} \int f(x) \frac{\eta}{x^2 + \eta^2} dx = f(0) \int_{-\infty}^{\infty} \frac{dt}{1 + t^2} = \pi f(0) \quad (2.4)$$

where we have used

$$\int_{-\infty}^{+\infty} dt \frac{1}{1 + t^2} = \arctan(+\infty) - \arctan(-\infty) = \pi.$$

Thus

$$\frac{\eta}{x^2 + \eta^2} \xrightarrow{\eta \rightarrow 0^+} \pi \delta(x). \quad (2.5)$$

3 Combined result (Sokhotski–Plemelj formula)

Putting the two limits together:

$$\frac{1}{x - i0^+} = \mathcal{P} \frac{1}{x} + i\pi \delta(x), \quad \frac{1}{x + i0^+} = \mathcal{P} \frac{1}{x} - i\pi \delta(x). \quad (3.1)$$

The spelling of the Polish name “Sochocki” has some alternatives, such as “Sokhotski”. In the physics literature this is the Dirac relation.